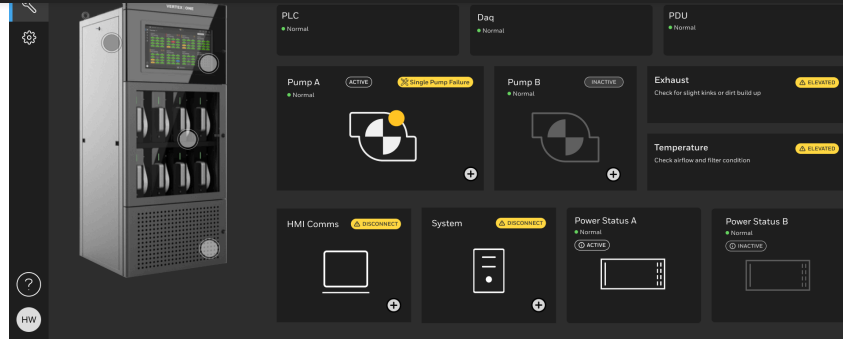


VertexONE - Gas Detection in Semiconductor FABs

As the UX Lead for the Vertex One program, I led the end-to-end design of Honeywell's next-generation toxic gas monitoring system for semiconductor fabs—an evolution of the award-winning Vertex Edge platform. The project integrated hardware, embedded interfaces, and cloud-ready HMI software to improve serviceability, ergonomics, and usability in high-containment environments.



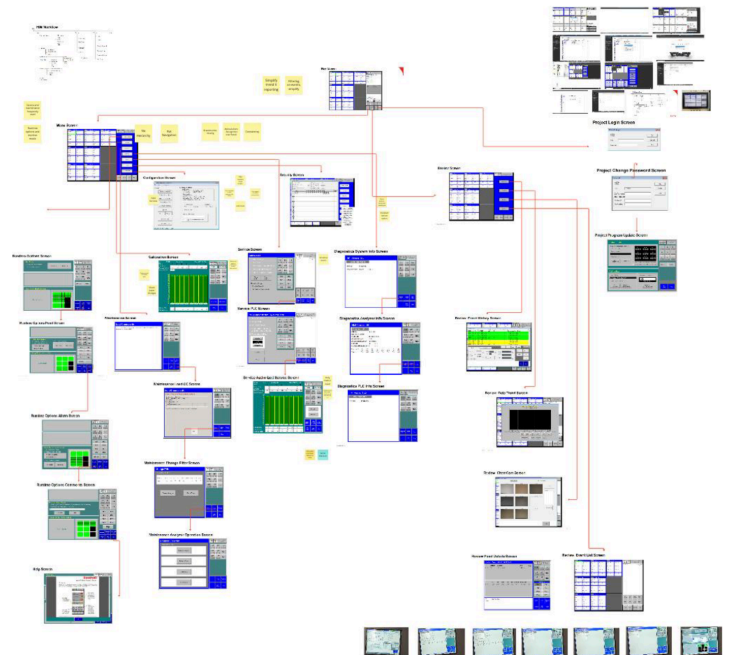


Challenge

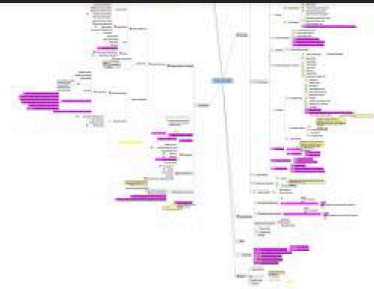
The core challenge involved designing the Human-Machine Interface (HMI) for Honeywell Vertex One, a critical industrial control system used for toxic gas detection in high-tech environments such as semiconductor fabs (e.g., Samsung Austin, Intel). In this environment, a gas leak is an invisible, fatal threat that demands split-second decisions and ensures near zero false positives.

The existing and legacy interfaces created significant usability and operational challenges for field service engineers (FSEs) and safety managers, including:

- **Data Overload:** Operators were overwhelmed by raw sensor data, making it difficult to maintain rapid situational awareness.
- **Legacy Workflows:** Maintenance workflows were complex and manual, contributing to lengthy repair times and potential operational downtime.
- **Ergonomics/Serviceability:** The physical design of the existing gas detection racks often made basic tasks like pump swapping and filter replacement difficult during long shifts



technical and difficult to follow, though ultimately very informative. I analyzed and translated these materials—ranging from specifications and process diagrams to test results—into clear, shareable frameworks and visual summaries. This helped cross-functional partners quickly grasp complex concepts and apply them effectively during design and development discussions.



Goals and Assumptions

My primary task as Lead UX Designer was to modernize and reimagine the Vertex One HMI, effectively creating a touch interface for toxic gas detection that allow field maintenance technicians to quickly diagnose problems within the fab and prevent a critical downtime or safety situation.

This involved three critical mandates:

1. **Reduce Cognitive Load:** Synthesize complex data (Parts Per Billion) into clear, immediate status alerts and actionable intelligence to support rapid decision-making.
2. **Bridge Hardware and Software:** Integrate the digital interface with the physical hardware and field maintenance service workflows (HMI/Human Factors Engineering).
3. **Future-Proof with AI/Data:** Design new features for predictive maintenance and diagnostics leveraging AI/ML insights to move operations from reactive response to proactive prevention

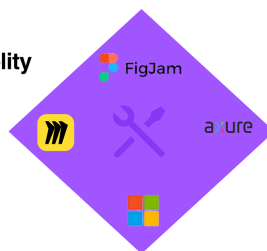
Discovery and Research

Interviews
Workshops
Synthesis
Competitive analysis



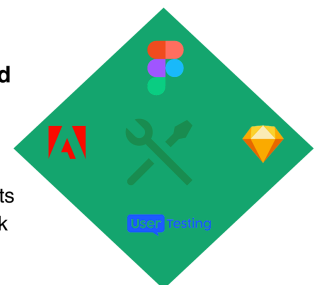
Low-Mid Fidelity and Testing

Storyboards
Site maps
Wireframes
Design Sprint



Prototyping and Document

Annotations
Dev handoff
Clickable user tests
Accessibility check



Research and Process

I undertook a comprehensive human-centered design approach including the definition of information architecture and navigation structures, competitive analysis, existing screen analysis, workshops,

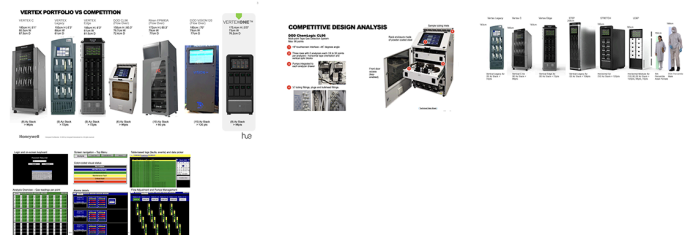
Minimizing Cognitive Load and Managing Data Overload

- Rigorously defined the information architecture and navigation structures to organize complex data flows for quick comprehension on both local touchscreens and remote management hubs.
- Prioritized clarity over aesthetics by implementing easy-to-read card-oriented layouts for touch-friendly navigation, minimizing cognitive overload.

- Proactively ensured accessibility by incorporating features like icon-assisted color coding to allow color-blind users to distinguish between status colors (green/red/yellow), recognizing the critical need for visual clarity in high-stakes environments.

Competitive Analysis, Existing Screen Analysis, Workflows

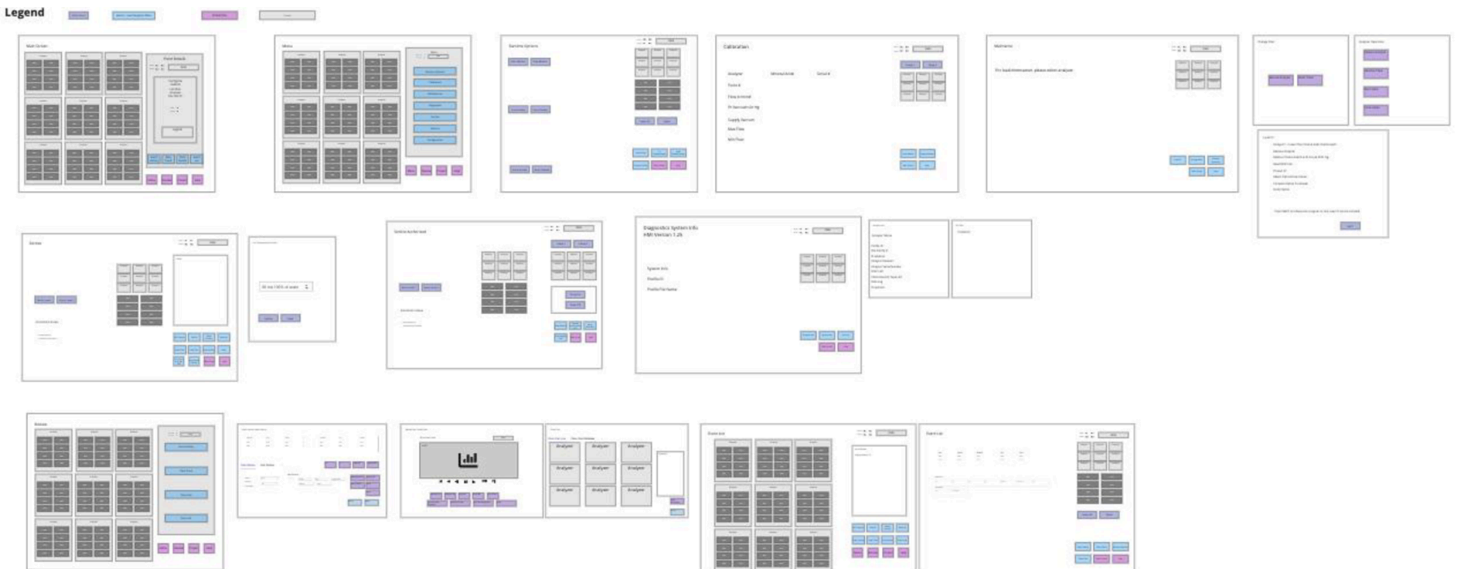
An evaluation of major competitors—such as Riken and DOD—revealed several usability and accessibility issues in their touchscreen interfaces. Poor color contrast, dense information layouts, and small touch targets make competitor screens difficult to operate and visually overwhelming.



Screen Analysis

Mapping the legacy Vertex screens helped to make sense of the decades old interface. Ahead of its time, the interface was designed prior to inventions like iPhone and iPad and the advent of touch interfaces.

Therefore, trial and error and a reliance on training manuals were prerequisites for usage. Wireframing the legacy interface aided my understanding of the product by removing unnecessary colors. Patterns and a system of thought began to emerge.



Workshop - Pre-work and Agenda

Held at Honeywell's Lincolnshire headquarters, the Vertex One UX Workshop brought together cross-functional stakeholders from product management, engineering, human factors, and field services to align on the next-generation Vertex One vision. The session focused on reimagining the human-machine interface (HMI) and overall user experience for toxic gas detection in high-risk semiconductor FAB environments.

As the UX Design Lead, I helped shape the workshop agenda, methods, and synthesis framework that guided subsequent concept validation at Samsung Austin. The workshop surfaced key challenges in integrating and maintaining Vertex systems, with emphasis on cybersecurity, preventative maintenance, documentation, system upgrades, and service efficiency. Discussions included field service team members for Intel, Samsung, and other partner accounts.

The technical support performance analysis revealed key factors affecting customer experience and service efficiency for the Vertex product line. The study identified documentation, software support, and analyzer troubleshooting as the top drivers of customer assistance requests.

Together, these patterns show that user friction peaks during installation, configuration, and maintenance workflows—especially around software compatibility, optics calibration, and documentation clarity. Streamlining these areas with improved procedural guidance, interactive digital tools, and enhanced remote diagnostics could significantly reduce escalation rates and technician downtime.

Quotes from transcript (Copilot Assisted)

[illegible]

We need something that allows us to remote into our systems specifically for troubleshooting purposes



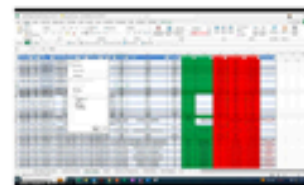
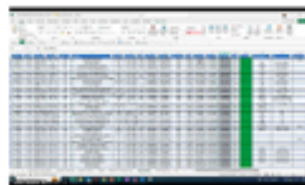
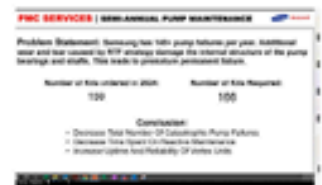
Ability to link into SBC using a Honeywell laptop for troubleshooting and loading of files/SiNs



Biggest struggles are related to SWS the PC and PLCs



Quarterly report for customers and notes

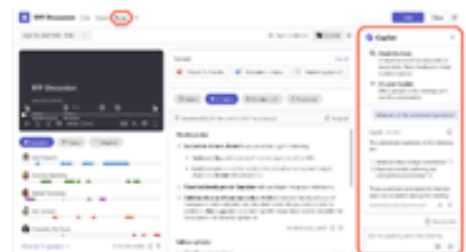


Meeting Notes - Assisted with MS Copilot

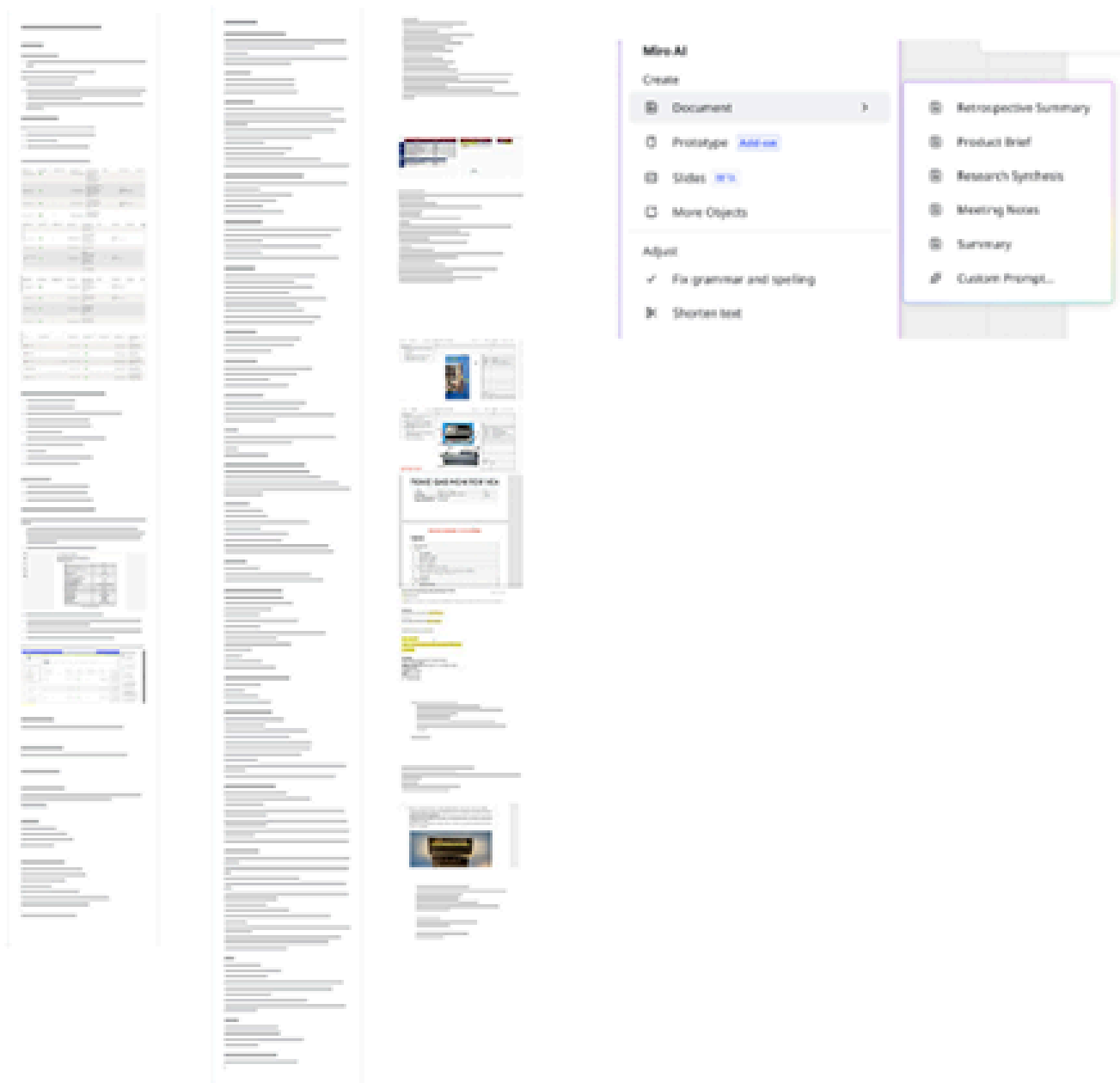
Meeting Notes - Assisted with MS Copilot

The meeting was held on [Date] at [Time] with [Participants]. The agenda included [Topics]. The meeting was productive and we discussed [Details].

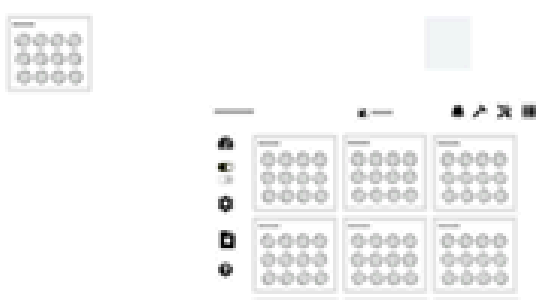
[Detailed meeting notes follow, including a list of action items and a summary of the discussion.]



(summarized with help from Miro AI docs)



Wireframes



How might we

Automate Reports?



How do we decrease point of fingers?



How do we get the customer to become more proactive



Improve Training? Combat turnover?



How can we reduce false alarms? Maintenance Faults?



How do we predict Failure. Predictive Maintenance



Chem Cassette Improvement



Show ECC and NDIR



Digital Maintenance Log



Parts replacement recommendations



Software releases comms improvement



PLC Wizard



Troubleshoot multiple Vertex from one screen



Duty Cycle Pump



Tape and Accuracy



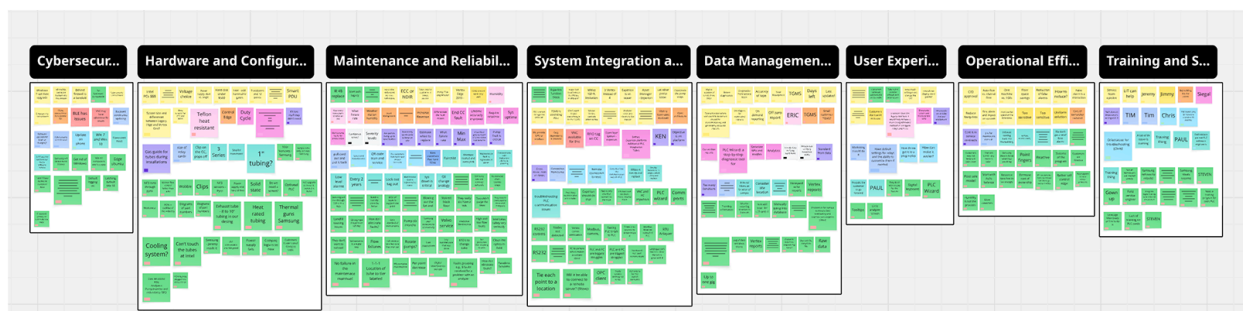
Clustering and AI synthesis

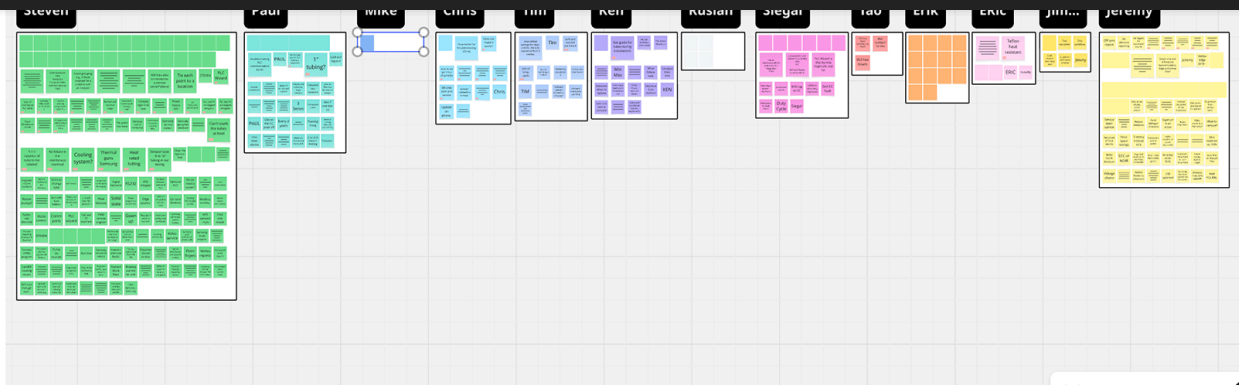
I organized all notes from the Day 1 and Day 2 workshops, first clustering them by the speaker's name. Then, I used AI to group the notes by keyword and sentiment. Each note is now tagged by both name and keyword, giving us richer context and faster access to insights. From this process, clear themes have begun to emerge, including cybersecurity, hardware and configuration, **maintenance and reliability**, system integration, data management, **user experience**, **operational efficiency**, and **training and support**.

These findings reinforce our initial pre-work and include new insights and anecdotes from FSEs explaining root causes. For example, flow rate fault alarms may result from CC wobble, while frequent downtime and support cases often stem from limited understanding of preventative maintenance.

We brainstormed recommendations such as in-app tips to address training gaps and **AI-driven custom guidance** based on environmental or location data. Additional ideas include improved **in-app documentation**, **diagnostics with recommended actions**, **clearer software versioning**, and **step-by-step guides** for optics cleaning.

Automated reporting could deliver tailored insights for different audiences, while remote troubleshooting and clearer Chemcassette visuals could reduce inefficiencies and miscommunication.





Theming - Patterns emerge

I gathered all information from workshops, tech support, meetings, and post-workshop debriefs, then sorted it into 20 themes. These would be pared down further to help us prioritize the best solutions and mitigate risk.

Themes identified during the analysis centered on **configuration/installation**, **maintenance**, and **user experience**. Configuration themes included software configuration, PLC configuration, upgrades, **installation**. Maintenance themes focused on documentation quality, **preventative maintenance**, spare parts management, valve performance, flow rate faults, optics issues and training. Finally, user-facing themes highlighted HMI improvements, **specific troubleshooting recommendations**, customer downtime and opportunities for Midas/Vertex integration.

With the workshop complete and key themes identified from pre-work with technical support tickets, stakeholder interviews, and the How Might We session, we now have clear priorities and initial solution paths to pursue. Themes such as **documentation**, **configuration**, **custom recommendations**, **preventative maintenance**, and **PLC issues** have emerged as core areas of focus. Next, we'll examine the note clusters around these themes to determine where to concentrate our efforts.



Synthesizing Top Themes and Prioritizing

With the help of AI and synthesizing tools, I'm able to identify the Top 5 themes that emerged from the workshop and interviews.

From there it is a matter of prioritizing features with stakeholders and identifying what to work and a strategy for the project. This identifies the features that we can control or can't control as well as which are the most impactful.

By identifying which solutions and features were high impact/high effort, high impact/low effort, low impact/low effort or low impact/high effort, the product team and I were able to confidently move forward with the knowledge that time was spent efficiently.

Top 5 Themes for each How Can We ...

How can we optimize portfolio?



Top 5 PORTFOLIO OPTIMIZATION Themes

- 1 **Industrial Design** - Family of Devices
- 2 **Resources** - Location of Engineers, Designers
- 3 **Research** - More VOC and diversify customer VOC
- 4 **Set up** - Register users and out of box self setup
- 5 **Sensor Improvement** - Smart Sensors, Reduce Size, Sensor Tech

How can we improve the docking experience?



Top 5 DOCKING STATION Themes

- 1 **Bump/CAL** - Faster, Modular, Innovate
- 2 **Gases** - More gases, Innovate Training, Auto Order Gas
- 3 **Industrial Design and UI** - Same base /universal Dock, Standardization, Updated Interface
- 4 **Engineering** - Platform, Minimal Components, Manufacturing
- 5 **Setup and Connect** - 1 click setup, simple connectors, Modular

How can we improve detector quality?



Top 5 DETECTOR QUALITY Themes

- 1 **Testing and QA Engineering** - Test with New Customers, Involve Sales, Thorough QA
- 2 **Sensors** - Size/Quality, R&D, Supply Chain Diversity
- 3 **Project Process** - Simplify Production, Document, RMA Portal
- 4 **Components** - Standard materials/screens, Off Shelf (not custom), Modular
- 5 **Connectivity** - Improve BLE, Modular, Pin connectivity, More connectivity options for all devices (wireless, Lora, satellite, cell)

Keyword Mentions



CONTEXT.

- Test with correct users
- Distributors are not users
- More VOC
- Smaller sensors
- Smart sensor tech
- Improve sensor quality/reliability
- Innovation resistance
- Language issues
- Quality issues
- Improve QA
- Test with new users
- Test with more users
- Standardize components
- Standardize UI
- Standard processes
- AR/VR training tech
- Consistent brand
- Bump/Cal Innovation
- Open connectivity
- Modular components
- Connect to HON products



KEY OPPORTUNITIES FOR IMPROVEMENT / INNOVATION

ENGAGE	SELL	INSTALL	USE	MAINTAIN	UPGRADE
Media and market portals require modernisation, tagging and reorganization	Loss of reputation due to reduced services such as education & consulting.	Improve usability, budget and production of education and training videos.	More investment/innovation in technology to stay ahead of competitors.	Make customer support easier to access and provide 24/7 support and more experts.	Leverage and utilize data collection while working cross functionally with other GBEs.
Cross discipline VOC plan need more rigor and structure. ☒	Complex quotation process.	We need to offer better technical support and consultation internal/external.	Portfolio is becoming dated because we are taking the safe route and not investing in R&D.	Right to repair, give customers more flexibility	More future forward planning for new product or update rollouts.
Honeywell, RAE and BW branding needs improved strategy and alignment. ☒	We need to repair and build our customer relationships as well as register end users.	Improvements are needed in having consistent, quality manuals and guides.	Disparity in size, shape, and design of portfolio is a major issue for a unified docking solution. ☒	Easier accessibility for sensor, pump and battery for replacement in field.	Build a better RMA portal.

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Short-term Solutions

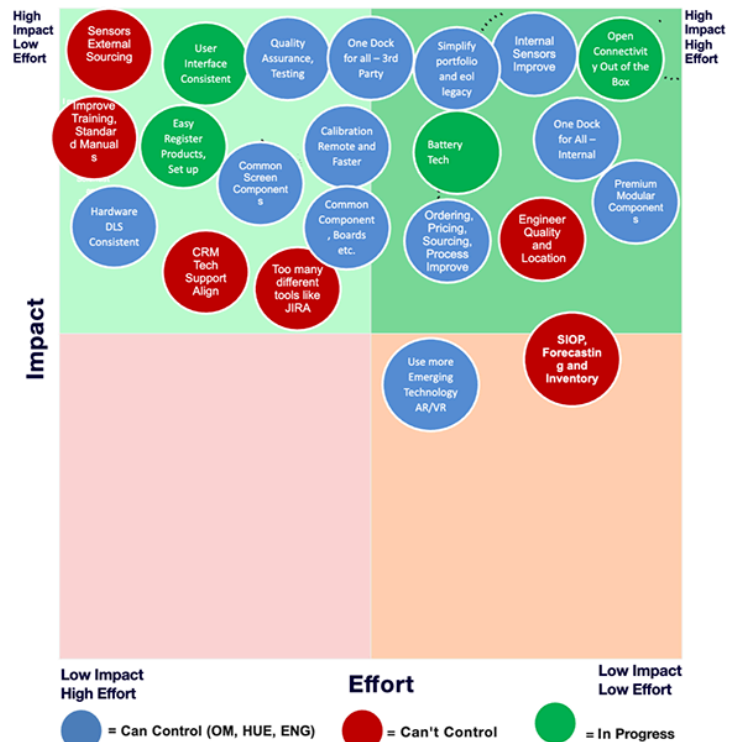
Least time, investment and/or effort

1. UI Consistency & Hardware DLS - Simplify & Consistent Interactions
2. Quality Assurance Improvement (More VOC)
3. Improve Setup and Registration of Products
4. Common Components - Screens, PCB Board
5. Improve Calibration - Remote Cal, Pre-Cal. Sensors and Cal. Speed

Long-time Solutions

Highest impact but require significant time, \$, effort

1. End of Life Devices to Simplify Portfolio
2. Internal Sensor Improvement (size, quality and reliability)
3. Universal Dock - One Dock for all devices
4. Battery Tech - smaller and longer charge
5. Open Connectivity - Easy connected setup out of box

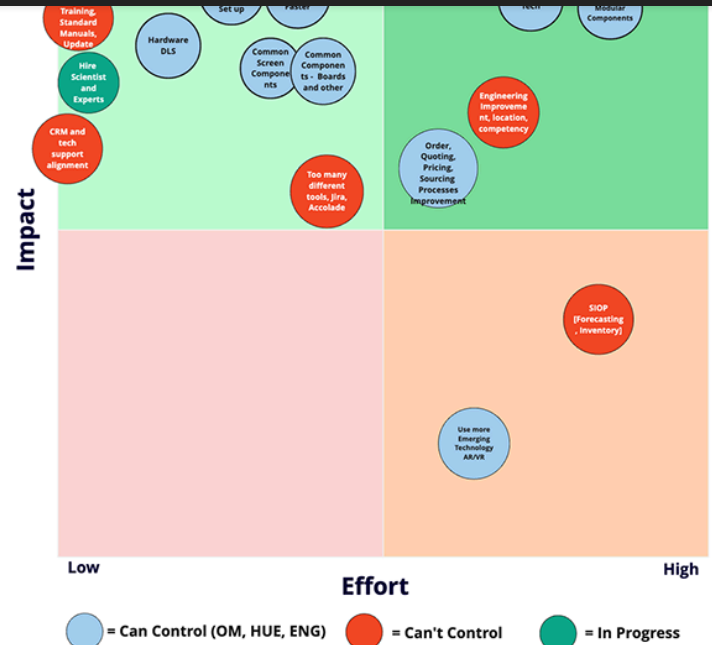


1. UI Consistency & Hardware DLS - Simplify Brands, Consistent Interactions
2. Quality Assurance Improvement (More VOC)
3. Improve Setup and Registration of Products
4. Common Components - Screens, PCB Board
5. Improve Calibration - Remote Cal, Self Cal. and Improve Cal. Speed

Top 5 - Impactful Solutions

(most time, money and/or effort)

1. End of Life Devices to Simplify Portfolio
2. Internal Sensor Improvement (size, quality and reliability)
3. Universal Dock - One Dock for all devices
4. Battery Tech - smaller and longer charge
5. Open Connectivity - Easy connected setup out of box



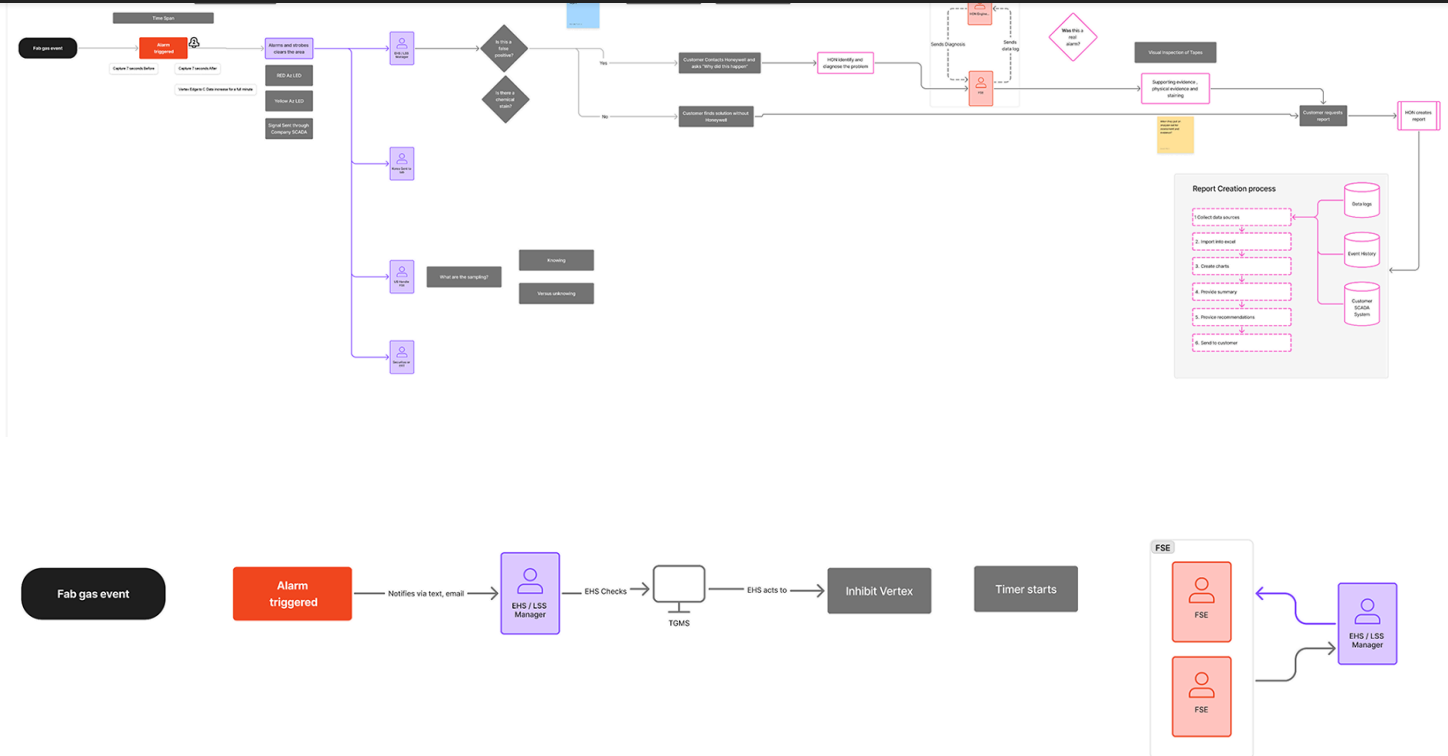
Top Risk and Mitigation - Strongest Solutions

(least time, money and/or effort)

1. UI Consistency & Hardware DLS - Simplify Brands, Consistent Interactions	
TOP RISK	MITIGATION
Brand Equity – RAE and BW	- Phase in and phase out, - Understanding of what we want to get done - Presentation to leadership (Real world feedback that we are washing out the RAE brand. Needs discussion.)
2. Quality Assurance Improvement	
TOP RISK	MITIGATION
Time and Cost	- Better Project Handoff Documentation - Streamline Project Process for true MVO releases - Organize and streamline tools (MS Teams, Sharepoint, JIRA, Confluence)
3. Improve Setup and Registration of Products	
TOP RISK	MITIGATION
Distributor Resistance	- Meet with distributors and explain benefits of end user registration of products - Improve project process to allow for extra time to work on non-features like setup - Allocate time for user testing
4. Common Components - Screens, PCB Board	
TOP RISK	MITIGATION
Engineering Resistance	- Leadership buy-in, then issue directive that engineers must find way to match designs - Financial analysis of cost savings from common screens and components - Create an engineering innovation team
5. Improve Calibration – Remote Calibration, Calibration Speed Improvement and Pre-calibrate sensors	
TOP RISK	MITIGATION
Lack of Investment	TBD

Documenting workflows and process

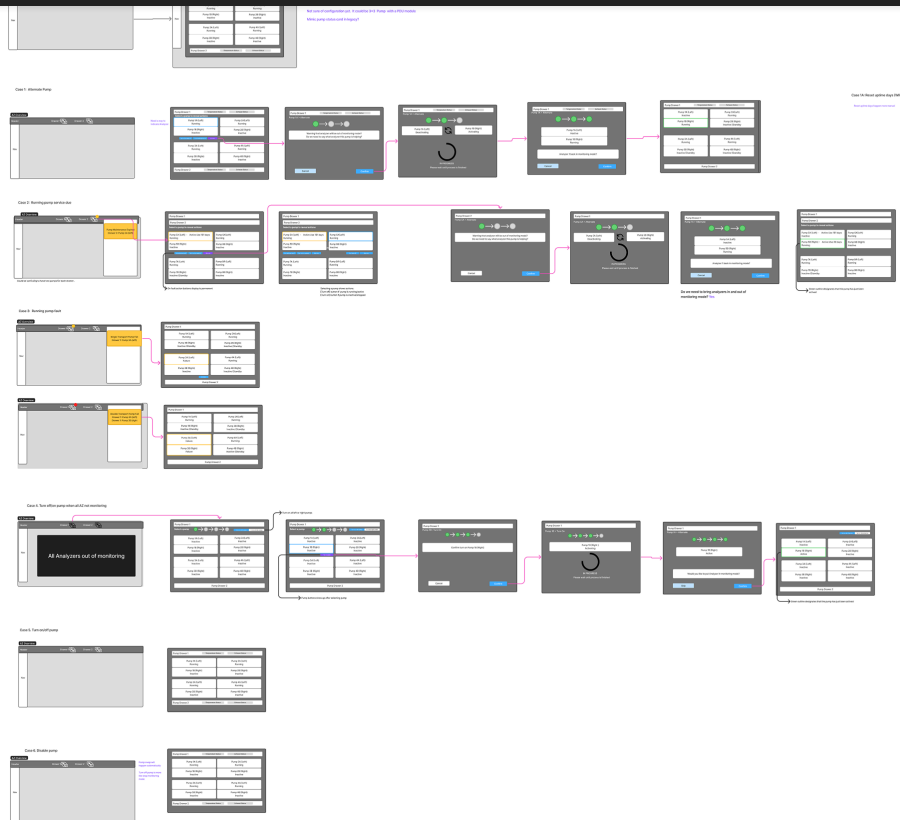
Mapping out flows with field service helped the team to better understand the steps involved when serious issues surrounding gas detection disrupted the FAB. Another key workflow introduced a **differentiating feature** for disabling auto pump swap and helped stakeholders visualize the functionality.



Wireframing and Usability Testing

In this case wireframing really helped to reduce the noise on highly complex concepts. The engineers and design team were better able to focus on the process and the placement of key elements as we designed a pump maintenance system.

By focusing solely on **structure**, **content hierarchy**, and the **step-by-step guidance** for complex procedures (such as optics cleaning and Chemcassette maintenance), we were able to quickly validate proposed workflows. This approach minimized early investment in high-fidelity visuals while maximizing team consensus on critical functionality like disabling auto pump swap—a key differentiating feature for field service efficiency.



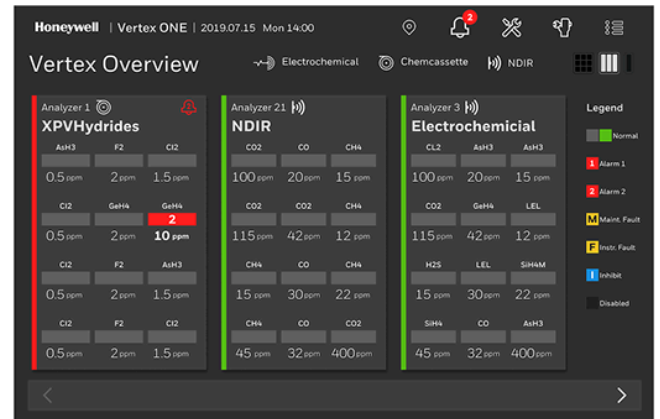
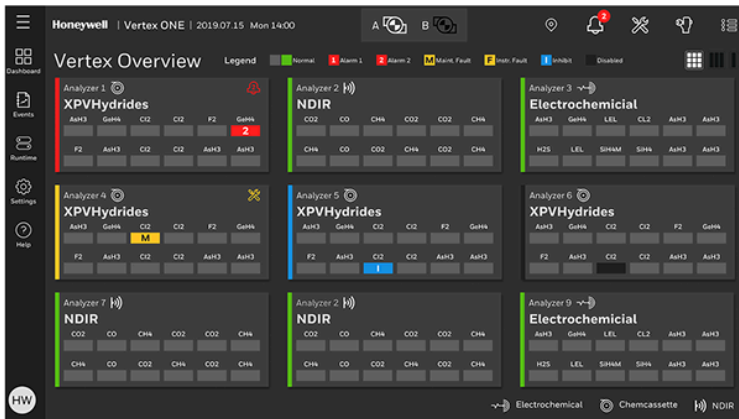
Usability testing with customers and internal stakeholders

As initial concepts crystalized the team, consisting of ID, research and UX, wanted to test out the validity with customer and stakeholders. Through this process we discovered that seeing green is calming and that focus mode color palette can mean something different within the staff at the fab. We tested step-by-step guidance, and the value of an AI assistant, all of which yielded surprising results. Lastly we wanted to test the value of analytics and how the visualization would be more valuable than the current table based event history.

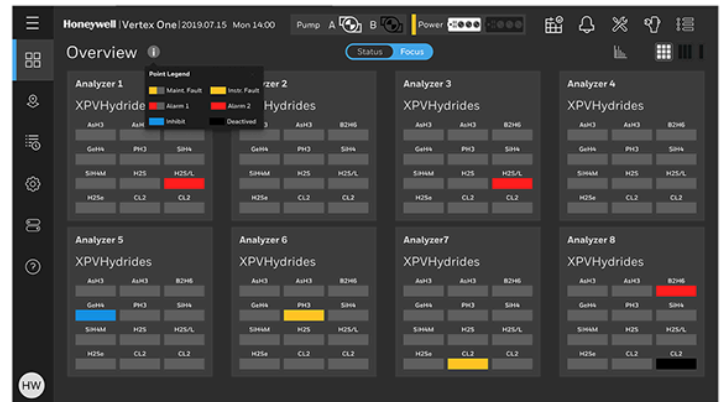
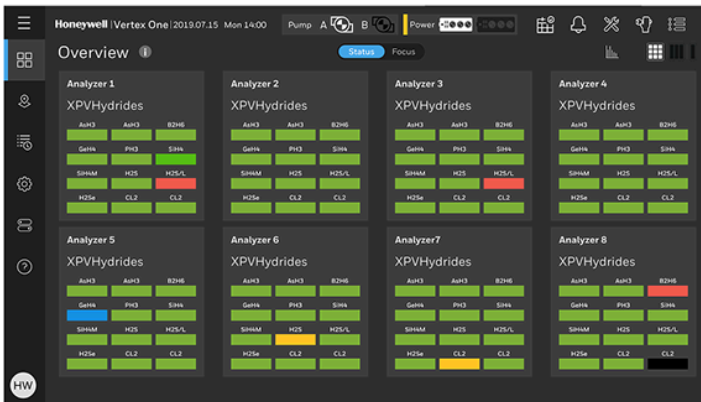
Matt McGlothlin



attention



Interaction and screen size preference preference? HMI, tablet, remote web interface



Overview Dashboard – Status Mode

✓ Full overview with green status indicators was preferred.



Optional Icons – Assist Mode

✓ Can be useful for training new hires, but would have to be disabled later

Overview Dashboard – Focus Mode

⚠ Gray points perceived as disabled or turned off. “Sleep mode” idea.
✓ Toggle option can help to switch modes based on user preference.

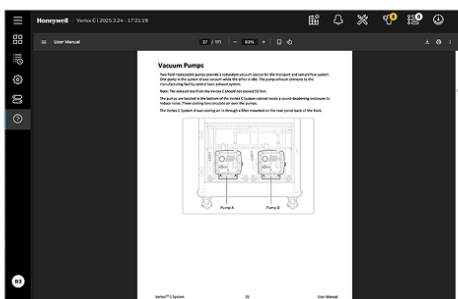
Status and Focus Toggle



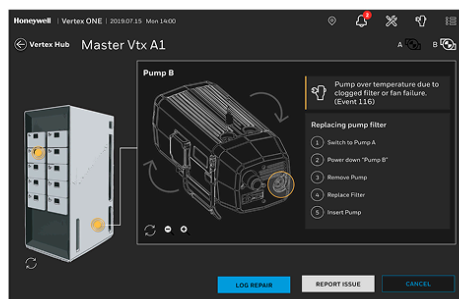
Need consistency in color representation across different interfaces

Color Indicators	Meaning
Green	Active
Red	Inactive or Faulty
Grayed Out	Disabled/Inhibited

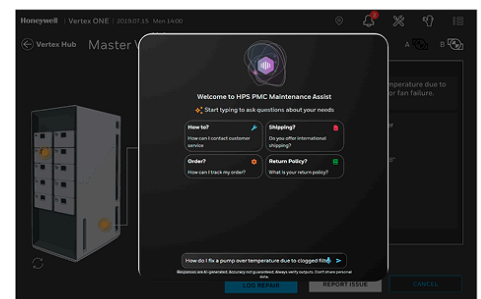
0 User Manual

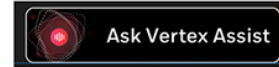
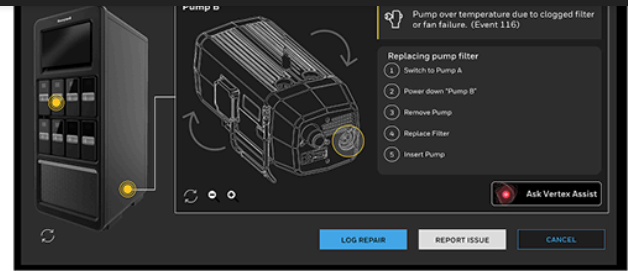


1 Visual step-by-guidance



2 Maintenance Assist on the Edge





Data analysis performed by HON Field Services for quarterly business reviews with SAS:

- Pareto chart for faults of all the HON equipment on site
- Types of faults
- Fault response time
- Uptime & downtime breakdown

Maintenance Assist on the Edge

⚠ Connectivity disabled for any Cloud-based solution

✅ Integration with Salesforce will add value to track LSS/FSE tickets

Designing for AI-Driven Prediction and Trust

- **Vertex Insights**, visualizing predictive maintenance based on machine learning from historical fault and alarm data, with the goal of preventing costly downtime.
- **Prediction Score** (Confidence Score) alongside the scenario and recommended action, recognizing that operators must understand the reliability of automated suggestions before taking action.
- **Vertex Assist** conversational AI assistant for the embedded touchscreen, enabling technicians to quickly ask natural language questions for troubleshooting help and maintenance schedules directly at the machine, diverting support tickets

ALARM/FAULT INTELLIGENCE

Analytics and insights with modern data visualization



Future Release

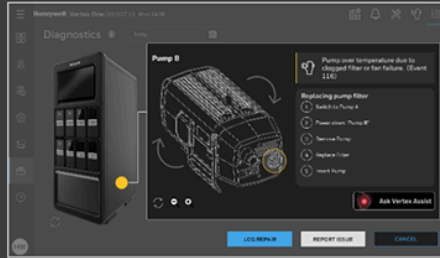
DOWNTIME PREVENTION

Predictive maintenance with AI-generated insights



SIMPLIFY LABOR INTENSIVE TASKS

Advanced diagnostics with guided steps and realistic diagram



FLEXIBILITY (MVO 2)

Remote management of multiple Vertex from various devices.



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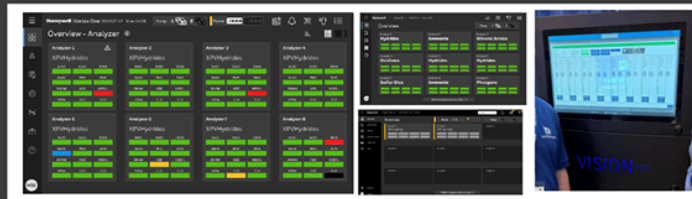
26

VERTEX ONE

Key HMI Enhancements

OVERALL EXPERIENCE

Continuous UI improvement (Edge > C > One) strengthens Honeywell usability advantage over major competitors like DoD.



USABILITY

Point mapping system increases productivity and improves recognition/recall of point location configurations.



INTEGRATION

TGMS inhibit timer enables FSE to prevent automatic faults



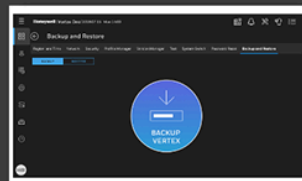
CYBERSECURITY

Auto-logout activates idle screen highlighting Vertex status.



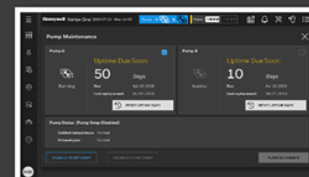
DATA INTEGRITY

Backup and restore prevents data loss and reduces downtime due to unforeseen conflicts.



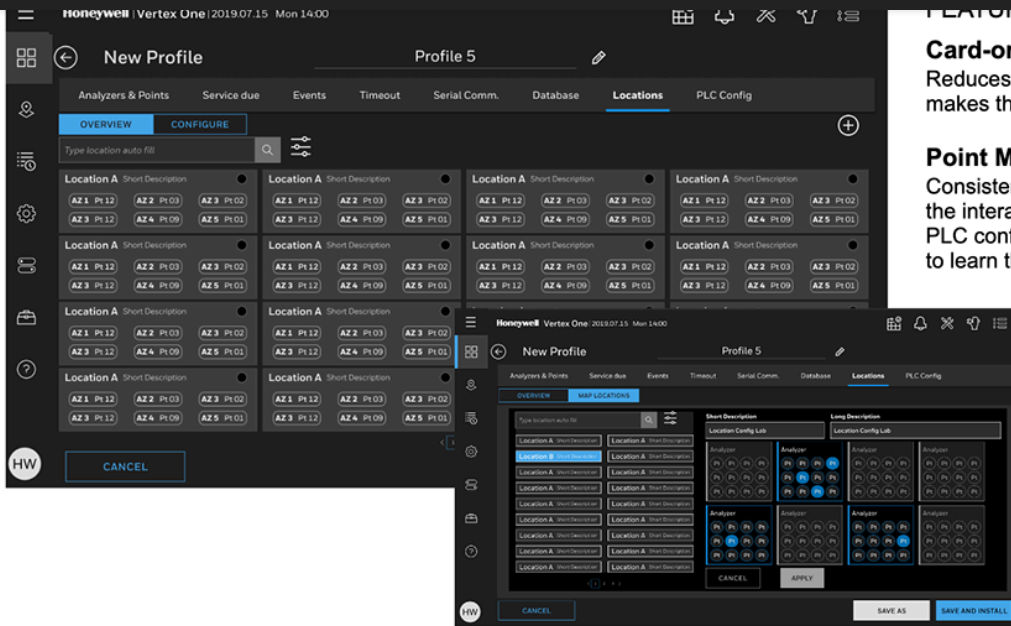
SAFETY

Disabling auto pump swap prevents potential safety issue when working on backup pump.



Honeywell

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Card-oriented Layout

Reduces table layout overload and makes the UI more touch friendly

Point Mapping

Consistency of point mapping since the interaction pattern is similar to PLC config. Therefore user only has to learn the functionality one time.

Search and Filter

Search and filter locations by short description, long description, AZ and PT

15

LOCATION GROUPING



FEATURES

Search, Sort and Filter

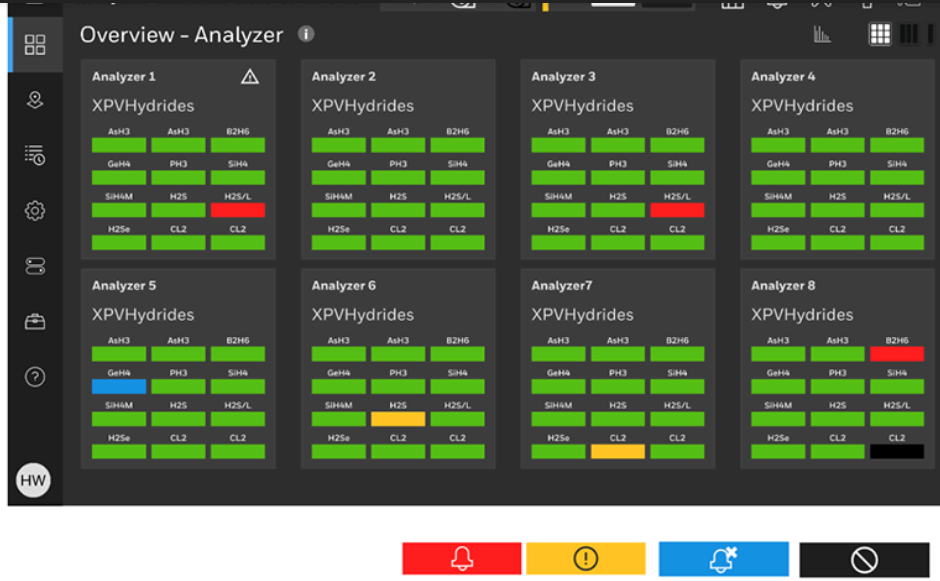
Search for a specific location or filter by alarm, fault, building. Sort by location name or location status

Point mapping

Each locations displays up to six points with Analyzer # and Point #

Gauge Charts

Location detail displays gauge chart with alarm thresholds, concentration and any faults highlighted in yellow.



Power Status

Header iconography allows user to monitor pump and power status in one glance. Selecting either graphic enables user to manage both.

Legend

New users unfamiliar with the UI can touch info icon for legend that defines point color and icons.

Layout Switcher

Switching between grid, column and single analyzer view gives users more flexibility and options for viewing analyzer status

Icon-assisted Points

Color blind users are able to distinguish between red and green easily with the help of icons

REMOTE MANAGEMENT (MVO2)

FEATURES

Multiple Vertex Overview

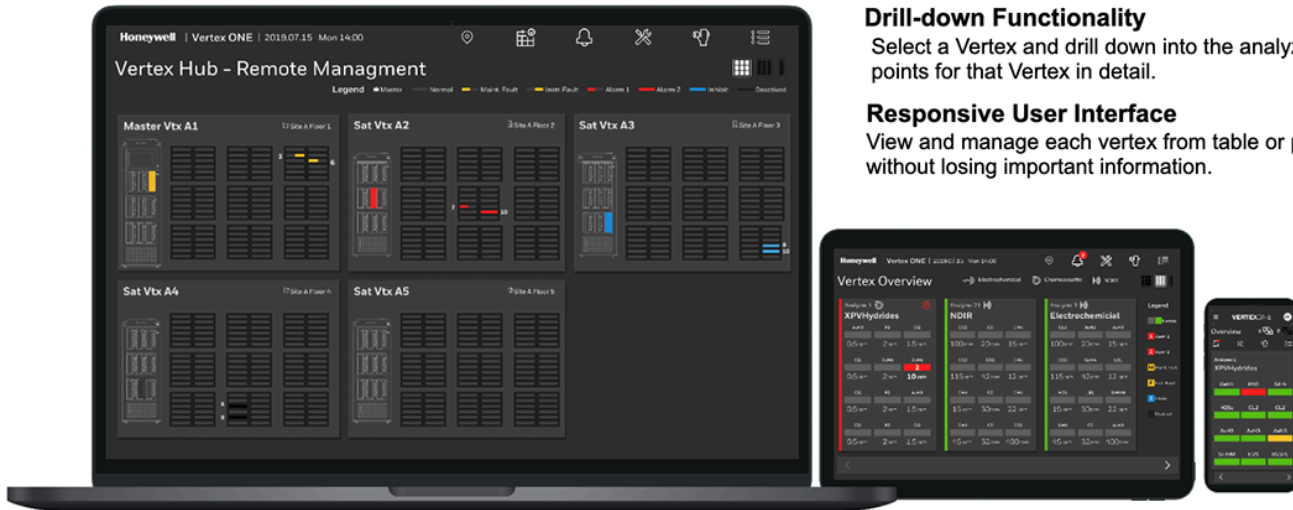
See status of all analyzers/points on all Vertex in one screen including alarms, faults and inhibits.

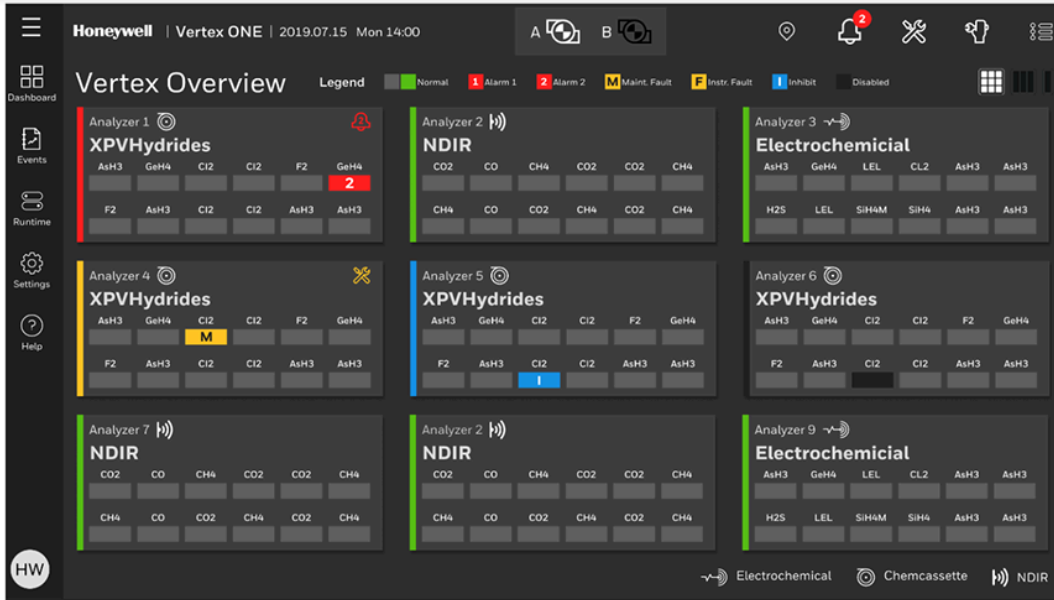
Drill-down Functionality

Select a Vertex and drill down into the analyzers/points for that Vertex in detail.

Responsive User Interface

View and manage each vertex from table or phone without losing important information.





Iconography

NDIR, Chemcassette and ECC icons signify to user which sensor analyzer contains

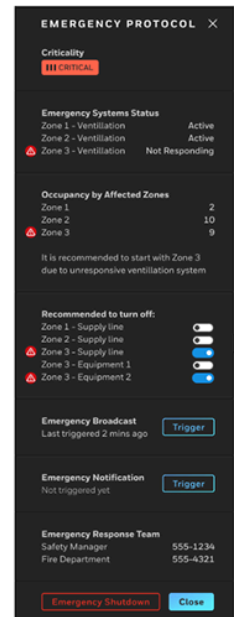
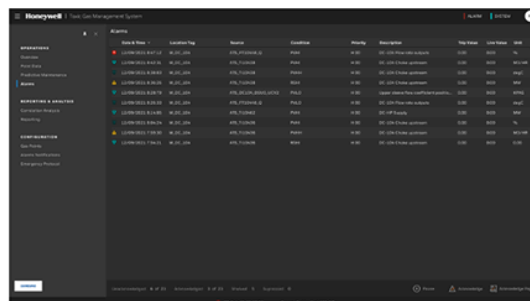
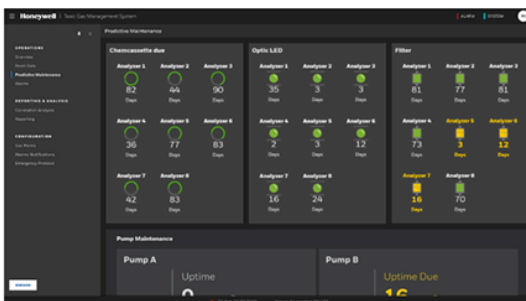
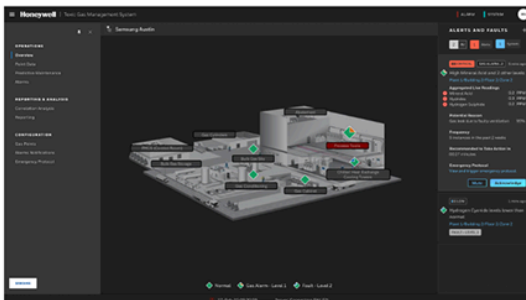
Legend

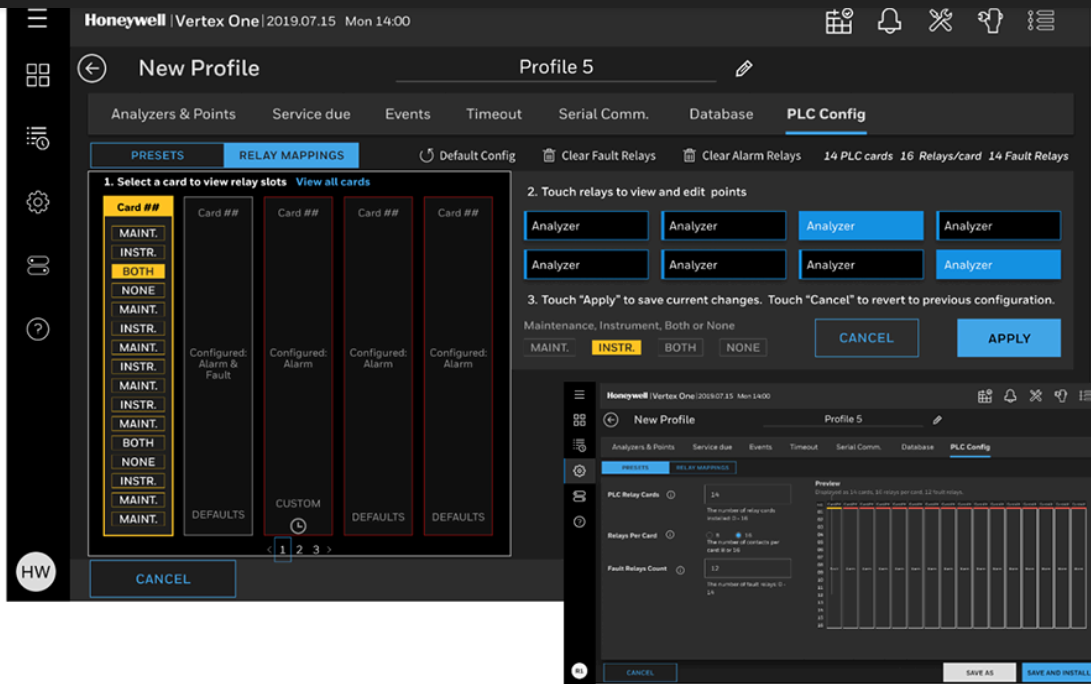
Bottom legend displays definitions for each type of sensor when selecting.

Sensor management

Bump and calibration process for ECC and NDIR sensors similar to Midas M.

TGMS INTEGRATION (WIP)





FEATURES

Presets and preview

User can visualize the card preset settings including type and amount as well as interact with tooltips that explain terminology = More efficiency.

Guided Relay Mapping

Each step is activated in a just-in-time fashion so that the user is never overwhelmed with choices.

Forgivable interface

Every step is reversible in case of mistake and changes are denoted with pending icons so user never loses track of changes.

Concepts, High Fidelity Prototypes and Dev Hand-off

After thoroughly and meticulously researching the new Vertex UI, synthesizing and theming the result.

After the interviews, VOC, usability studies, prototypes and wireframing ... only then am I ready for finalizing the user interface. This feels like the easy part after all of the preparation. My method is for user interface design relies on research, screen comparisons, and brand considerations. I want to use the latest design system that the Vertex seems modern and there is also special consideration for the touch screen nature of the HMI and the fact that people may be looking up at the interface. I considered alarms, complicated maintenance scenarios and the latest trends in AI and digital twins.

Concepts: Overview screens and event panels

The new scan line mirrors the exterior indication of alarms and alerts. The icons within alarm and faults on each point is a special consideration for low vision and color blind users as red and green is a common challenge for color blind users. We made the pump status more easy to digest and more descriptive of faults since this is one of the differentiators the new Vertex. Location grouping was a submitted invention disclosure for this project as was another feature for disabling the automatic pump swap.

Matt McGlothlin



Analyzer 1

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 2

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 1

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 4

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 1

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 6

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 7

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 8

XPV Hydrides

?

HW

Service Due

Vertex One 2025.07.15 Mon 14:00

PUMP STATUS: A ACTIVE B INACTIVE

Overview

ANALYZER VIEW LOCATION VIEW

Analyzer 1

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 2

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 1

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 4

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 1

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 6

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 7

XPV Hydrides

SiH4	SiH4	SiH4
SiH4	SiH4	H2S
PH3	PH3	CI2
H2Se	H2Se	H2Se

Analyzer 8

XPV Hydrides

?

Service Due

Alarm 1

Location Name

H2S 8.01 ppm

22.07.2019 1:31:20 PM

Analyzer 1 > Pt 01

Acknowledge Reset

Alarm 2

Location Name

H2S 3.563 ppm

22.07.2019 1:31:20 PM

Analyzer 1 > Pt 01

Acknowledge Reset

Flow rate low dasfsf...

F101

Analyzer 1 > Pt 01

22.07.2019 1:31:20 PM

Location Name

Acknowledge Reset

Optic Noise

F101

Analyzer 1 > Pt 01

22.07.2019 1:31:20 PM

Location Name

Acknowledge Reset

Optic Noise

F101

Analyzer 1 > Pt 01

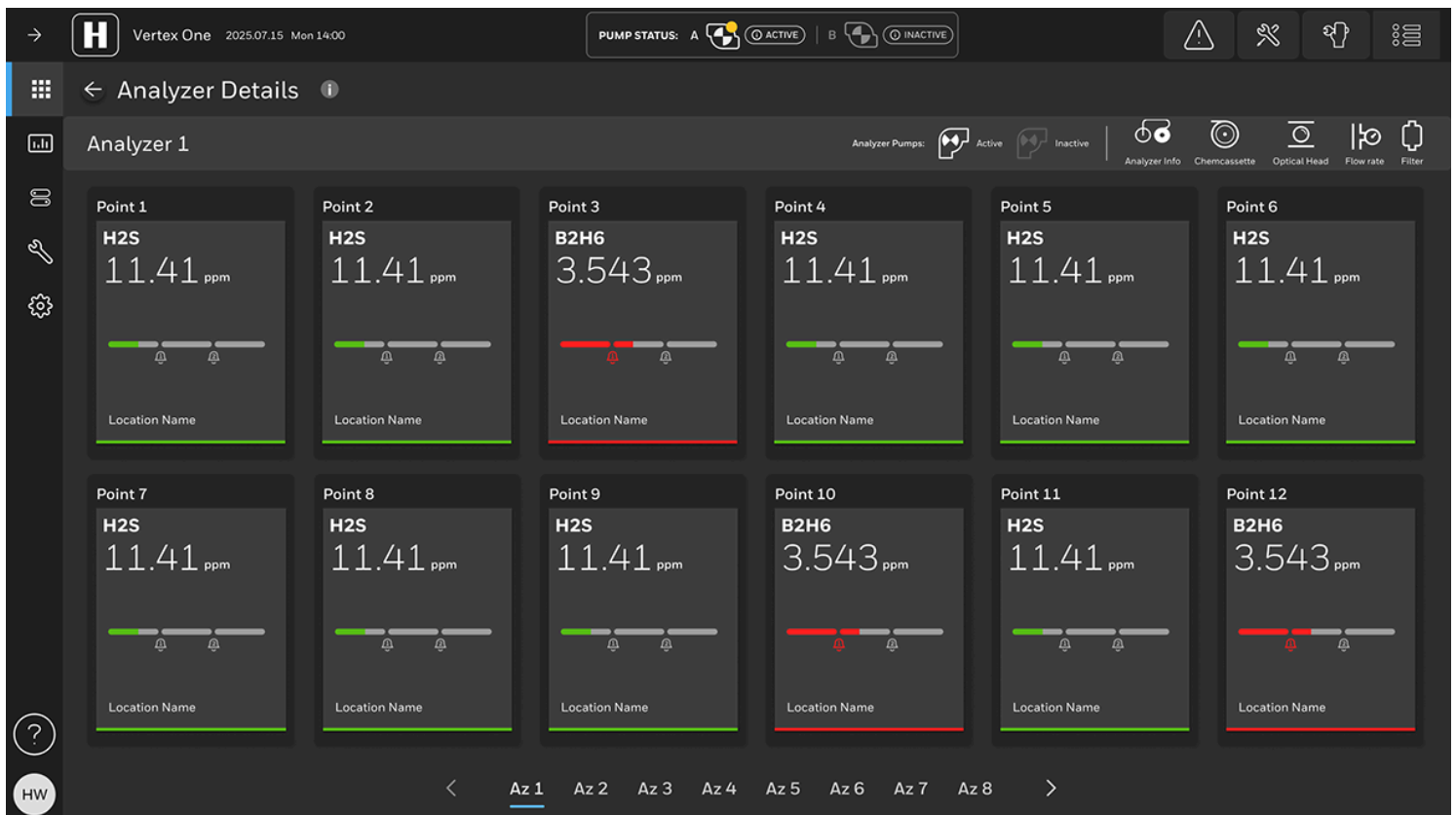
22.07.2019 1:31:20 PM

Location Name

Acknowledge Reset

< 1 2 ... # >

Acknowledge All Reset All



Concept: PLC touch configurator

stadium when purchasing a ticket and mapping that to the number of people attending. What options do they want with for their seats and who should be notified.

→

H

Vertex One 2025.07.15 Mon 14:00

←

New Profile

ⓘ

Analyzer Details

Service Due

Events

Timeouts

Serial Comm

Database

PLC Config

Location

Presets

Fault Relay Mappings

Alarm Relay Mappings

	Card 04	Card 05	Card 06	Card 07
Slot 01	MAINT	AL2	AL2	AL2
Slot 02	BOTH	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL2	AL1	AL1	AL1
	Card 08	Card 09	Card 10	Card 11
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
	Card 12	Card 13	Card 14	Card 15
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

4-15 of 45 items

4-15
16-27
28-39
40-45

Card

Card

Nothing selected yet

Select a card relay first

Cancel

Save As

Install

Save

→

H

Vertex One 2025.07.15 Mon 14:00

←

New Profile

ⓘ

Analyzer Details

Service Due

Events

Timeouts

Serial Comm

Database

PLC Config

Location

Presets

Fault Relay Mappings

Alarm Relay Mappings

	Card 01	Card 02	Card 03	Card 04
Slot 01	MAINT	MAINT	MAINT	MAINT
Slot 02	INSTR	INSTR	INSTR	BOTH
Slot 04	MAINT	MAINT	MAINT	AL1
Slot 04	INSTR	INSTR	INSTR	AL2

Select Fault Type *

☐ Maintenance Fault
☐ Instrument Fault
☐ Both
☐ None

Cancel

Apply

Select analyzer points to map

☐ Analyzer 01
☐ Analyzer 02
☐ Analyzer 03
☐ Analyzer 04

☐ Analyzer 05
☐ Analyzer 06
☐ Analyzer 07
☐ Analyzer 08

Cancel

Save As

Install

Save

Analyzer Details Service Due Events Timeouts Serial Comm Database PLC Config Location

Presets **Fault Relay Mappings** Alarm Relay Mappings

	Card 01	Card 02	Card 03	Card 04
Slot 01	MAINT	MAINT	MAINT	MAINT
Slot 02	INSTR	INSTR	INSTR	BOTH
Slot 04	MAINT	MAINT	MAINT	AL1
Slot 04	INSTR	INSTR	INSTR	AL2

Select analyzer points to map

- ☒ Analyzer 01
 ☐ Analyzer 02
 ☒ Analyzer 03
 ☐ Analyzer 04
☐ Analyzer 05
 ☐ Analyzer 06
 ☒ Analyzer 07
 ☐ Analyzer 08

Select Fault Type *

- ☒ Maintenance Fault
 ☐ Instrument Fault
 ☐ Both
 ☐ None

Cancel

Apply

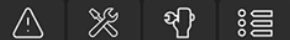
Cancel

Save As

Install

Save

→ **H** Vertex One 2025.07.15 Mon 14:00



← New Profile ⓘ

Analyzer Details Service Due Events Timeouts Serial Comm Database PLC Config Location

Presets **Fault Relay Mappings** Alarm Relay Mappings

	Card 04	Card 05	Card 06	Card 07
Slot 01	MAINT	AL2	AL2	AL2
Slot 02	BOTH	AL2	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL2	AL1	AL1	AL1

	Card 08	Card 09	Card 10	Card 11
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

	Card 12	Card 13	Card 14	Card 15
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

4-15 of 45 items < 4-15 16-27 28-39 40-45 >

Select analyzer points to map

Analyzer 01

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 02

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 03

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 04

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 05

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 06

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 07

01	02	03	04
05	06	07	08
09	10	11	12

Analyzer 08

01	02	03	04
05	06	07	08
09	10	11	12

Select Alarm Type *

- ☐ Alarm 1
 ☒ Alarm 2
 ☐ Both
 ☐ None

Cancel

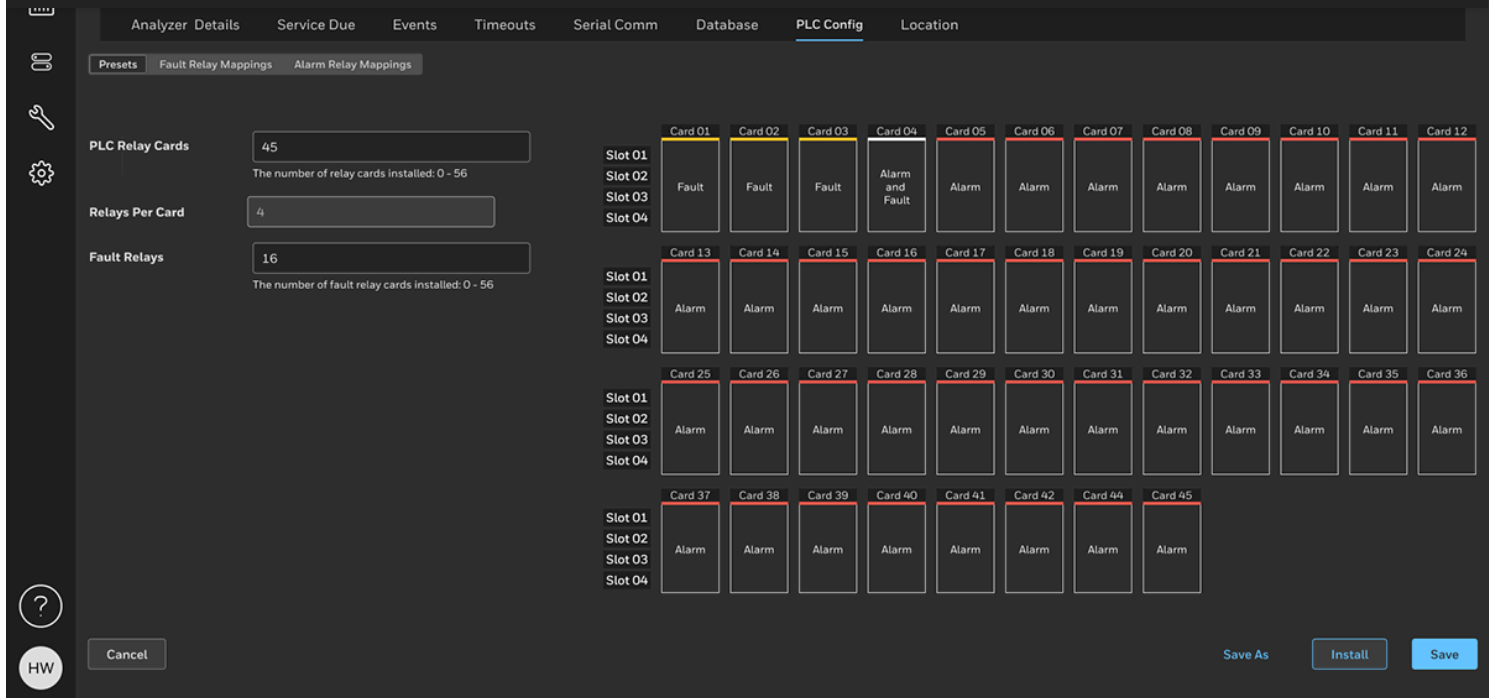
Apply

Cancel

Save

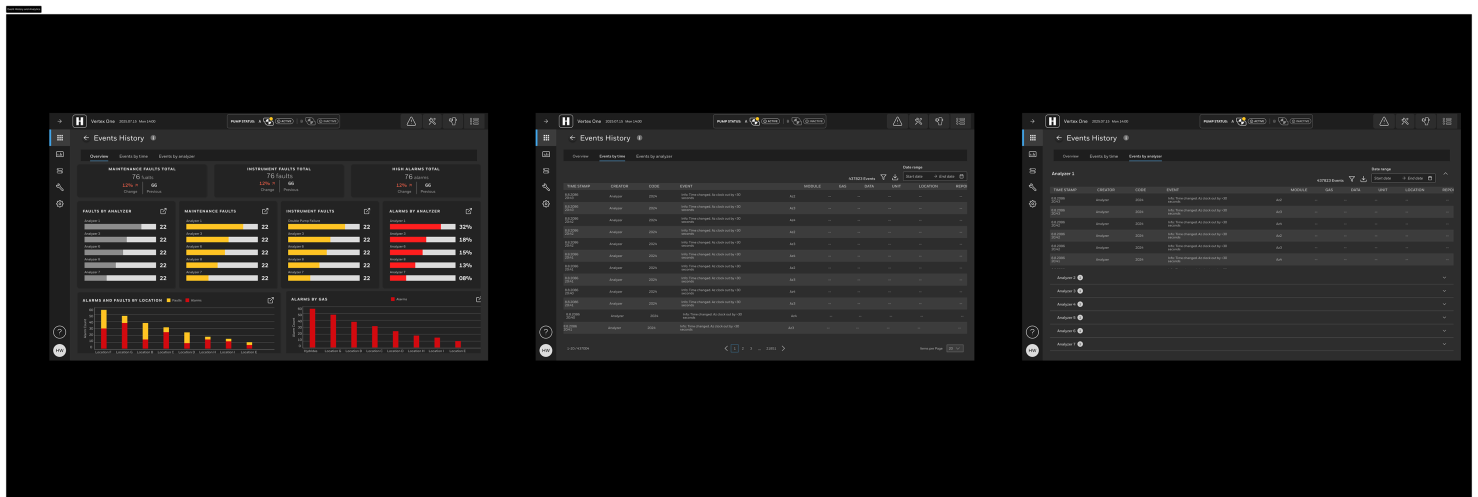
Save As

Install



Concept: Analytics and Event History

We also heard that field service technicians wanted more insights into faults and alarms on the Vertex. So we provided easy to digest visualizations that also helped with the quarterly reporting process for customers. The resulting savings could amount to \$1M+ annually and increase revenue for Honeywell in the high tech space.



Concept: Training and Guided Maintenance

We heard repetitively that documentation was lacking and that training was needed. We solved this by explicitly detailing the steps of complex tasks like replacing Chemcassettes while providing instant and



Runtime Operation ?



Analyzer Monitor Point Runtime Alarm Inhibit Analyzer Operation Bulk replace Chemcassette



HW

☐ Analyzer 1

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 2

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 3

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 4

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 5

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 6

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 7

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

☐ Analyzer 8

90 Days left

5.2.2026
Expiry date

Gas family: XPV Hydrides

Cancel

Replace Chemcassettes

Replace Chemcassettes



Preparing by opening optic gates...

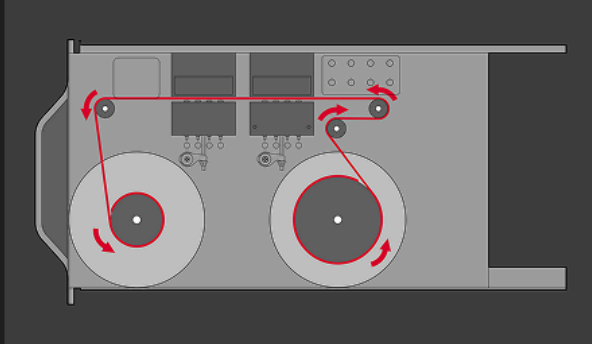
- Analyzer 1 Optic gate is opened
- Analyzer 3 optic gate is opened
- Analyzer 5 is optic gate is opened
- Analyzer 7 is optic gate is opened
- Analyzer 8 optic gate is opened



- Release analyzers
- Replace Chemcassette
- Uploading data from RFID
- Verify Chemcassette status
- Adjust white, light/dark gray
- Resume monitor or idle



Please open each analyzer and replace the chemcassette.
When the all chemcassettes are placed, please click NEXT.



Cancel

Next

Replace Chemcassette: Analyzer 1



Uploading data from the chemcassettes RFID tag



- ✓ Release analyzer
- ✓ Replace Chemcassette
- Uploading data from RFID
- Verify Chemcassette status
- Adjust white, light/dark gray
- Resume monitor or idle



Information of the new chemcassettes

Analyzer 1

Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 7

Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 3

Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 8

Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 5

Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

- ✓ Release analyzer
- ✓ Replace Chemcassette
- ✓ Uploading data from RFID
- Verify Chemcassette status
- Adjust white, light/dark gray
- Resume monitor or idle

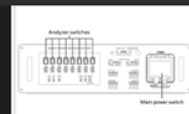
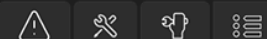
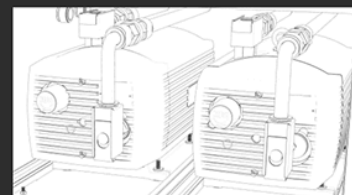
Cancel

Next

Concept: Diagnostics and Recommendations

Ideas circulated around specific recommendations with the rise of artificial intelligence. Internal stakeholders loved this feature including FSEs with many years of experience. The recommendations synthesized from various data sources could be a game changer for Honeywell and allow technicians to spend more time with the customer and less time figuring out how to troubleshoot edge case issues. The digital twin aspect of the diagnostics section received applause from sales, marketing and technicians.

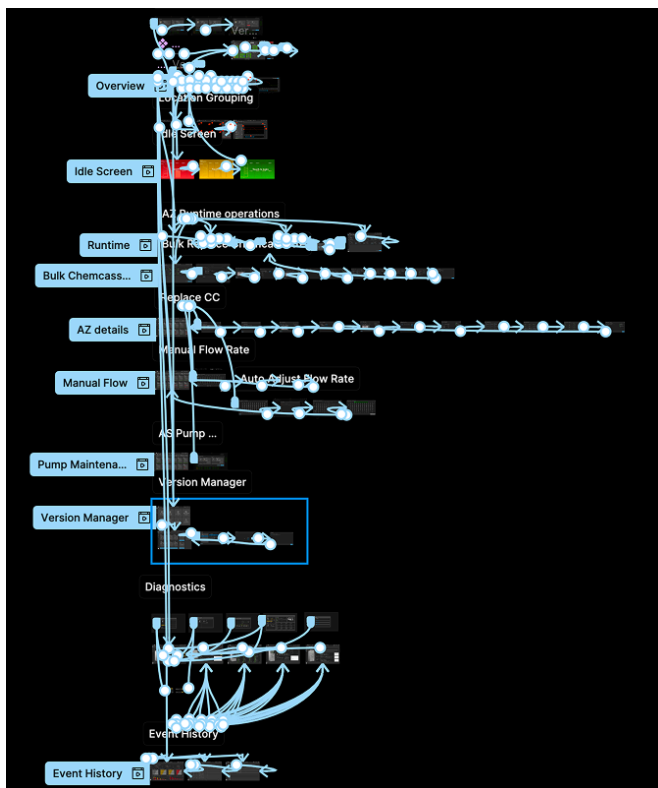
"With the analytics and diagnostics ... This is something we can sell" - Honeywell High Tech President



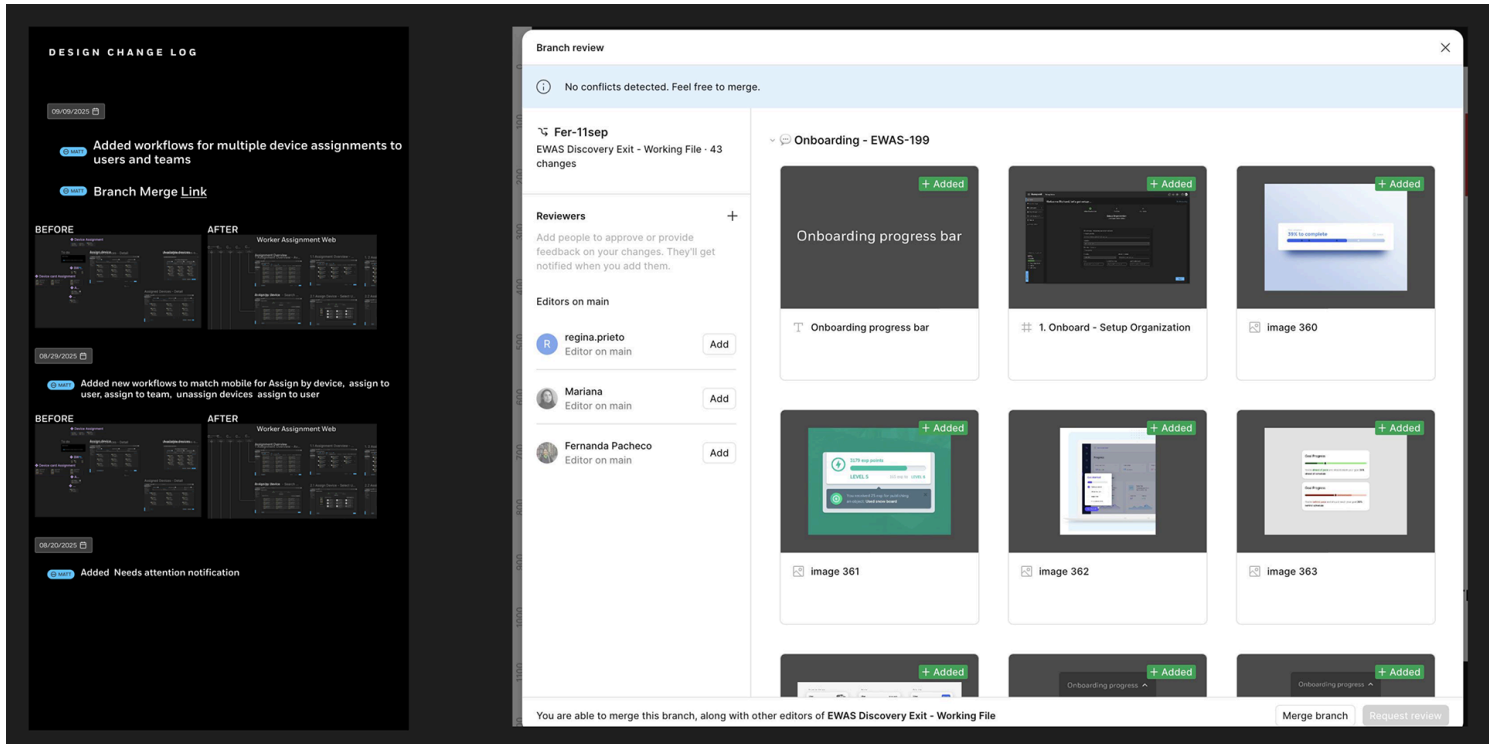


High Fidelity Prototype

The prototype helped to test the final screens with users and internal stakeholders alike. Every screen was linked up to the prototype and we were able to hook it into a tablet that mimicked the actual Vertex machine.



During the dev handoff process it's critical to keep designs updated when there will inevitable be minor tweaks. This is why I implemented a versioning system that included branch and merge functionality similar to GitHub. This native-Figma functionality allowed designers and developers to stay in sync and schedule updates after release in subsequent sprints. Annotating designs is another critical and often overlooked part of the design process. Annotations reduce the amount of back and forth between developers and stakeholders alike.



Outcomes

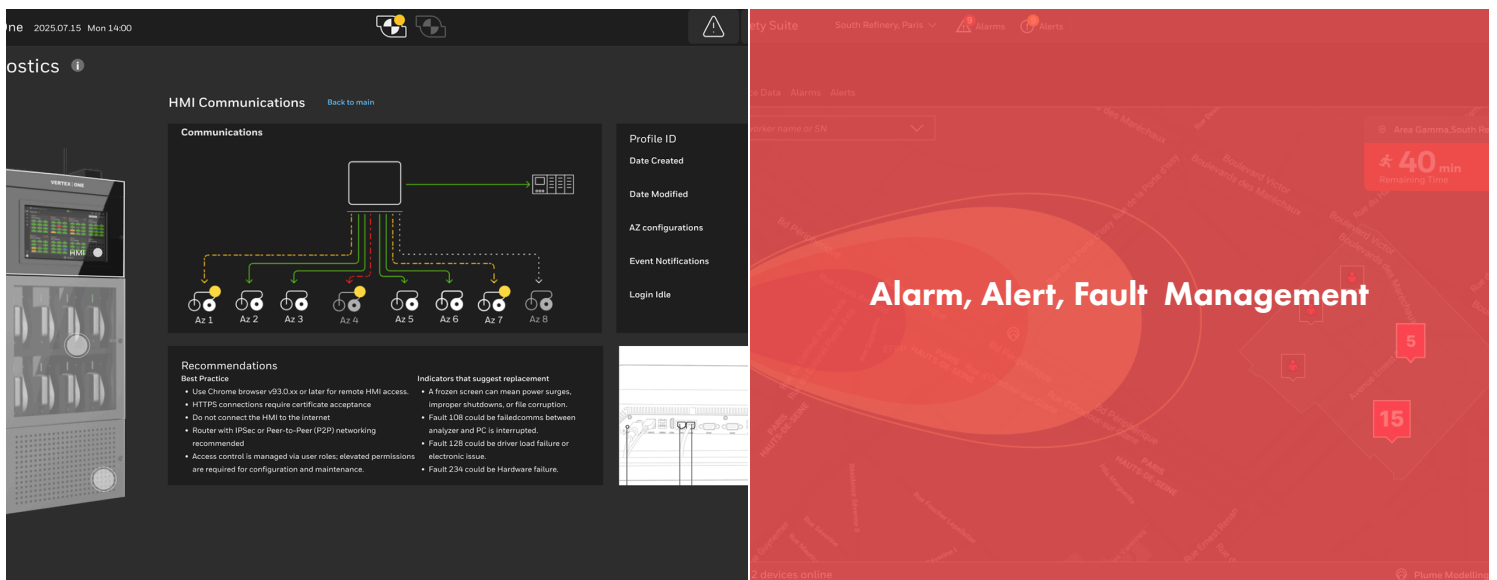
The design actions for Vertex One led to an HMI solution recognized for its quality, successfully transforming a fragmented system into a cohesive, mission-critical interface, positioning it as a significant competitive advantage for Honeywell:

- **Mission Assurance/Safety:** The design directly enhanced safety by introducing features like Pump Inhibit Mode and Auto-Logout Idle Screens to prevent unintended faults or unsafe operations during maintenance activities, emphasizing safety over convenience.
- **Efficiency & Usability:** The intuitive UI, combined with enhanced diagnostics and remote management concepts (Vertex Hub), significantly reduced the physical strain on technicians (less walking to machines) and streamlined labor-intensive service tasks.
- **Design Governance:** The specialized design system for the Vertex touchscreen interface utilized reusable components and placeholders. This allowed the design team to establish consistency with Honeywell's larger enterprise design standards as well as accomodate flexible and scalable upgrades.

new vertex = scalability with future releases incorporating premium features that integrate with even larger systems, and that include AI-based features to automate maintenance.

Presentations

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